



DATA-DRIVEN ANALYSIS OF SKIN LESIONS: A SQL-BASED APPROACH TO SUPPORTING EARLY SKIN CANCER DETECTION: Insights for DermAI Diagnostics.

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INTRODUCTION

- ❖ Cancer remains one of the leading causes of illness and death worldwide, with **18.1 million new cases and 10 million deaths recorded in 2020**, according to global cancer surveillance reports.
- ❖ Within this broader landscape, **skin cancer has emerged as one of the fastest-growing and most common cancers globally**. In 2022, melanoma alone accounted for **over 331,000 new cases**, ranking among the most frequently diagnosed cancers worldwide.
- ❖ DermAI Diagnostics focuses on addressing this growing challenge by integrating clinical dermatology data with AI-driven analysis.
- ❖ This project analyses patients' demographics, environmental exposures, and lesion characteristics to support early detection, improve diagnostic accuracy, and generate structured, SQL-ready data for research, training, and machine learning.



PROBLEM STATEMENT

Global Skin Cancer Challenge

❖ Skin cancer remains a critical health issue with severe outcomes when diagnosis is delayed.

Data-Driven Early Detection

❖ SQL queries help uncover hidden patterns in clinical data to support earlier skin cancer diagnosis.

Environmental and Demographic Factors

❖ Examining risk factors like pesticide exposure and demographics can improve understanding of lesion development.

Clinical Decision Support

❖ Insights from data analysis aim to enhance clinical decisions and machine learning model preparation.



OBJECTIVES

- ❖ Build a structured SQL database for analyzing clinical and lesion data.
- ❖ Identify demographic and environmental factors linked to different skin lesion types.
- ❖ Examine lesion characteristics to distinguish cancerous from benign patterns.
- ❖ Prepare a machine-learning-ready dataset for AI-based early detection.
- ❖ Support dermatological and epidemiological research with high-quality, organized data.





DATA DESCRIPTION

Patient_Info Table Overview

Includes 13 fields covering lifestyle, environmental exposures, and household infrastructure to assess risk factors.

Lesion_Info Table Details

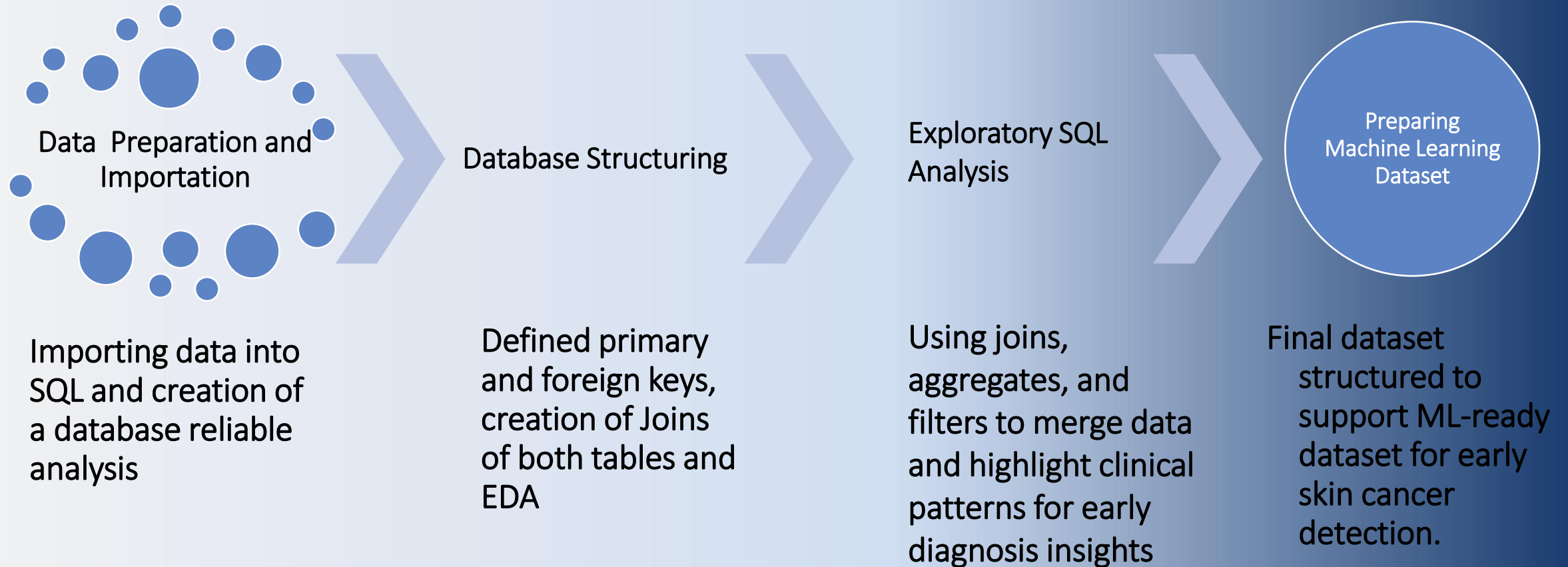
Contains 16 clinical lesion characteristics including measurements, lesion types, symptoms, and skin classifications.

Tool

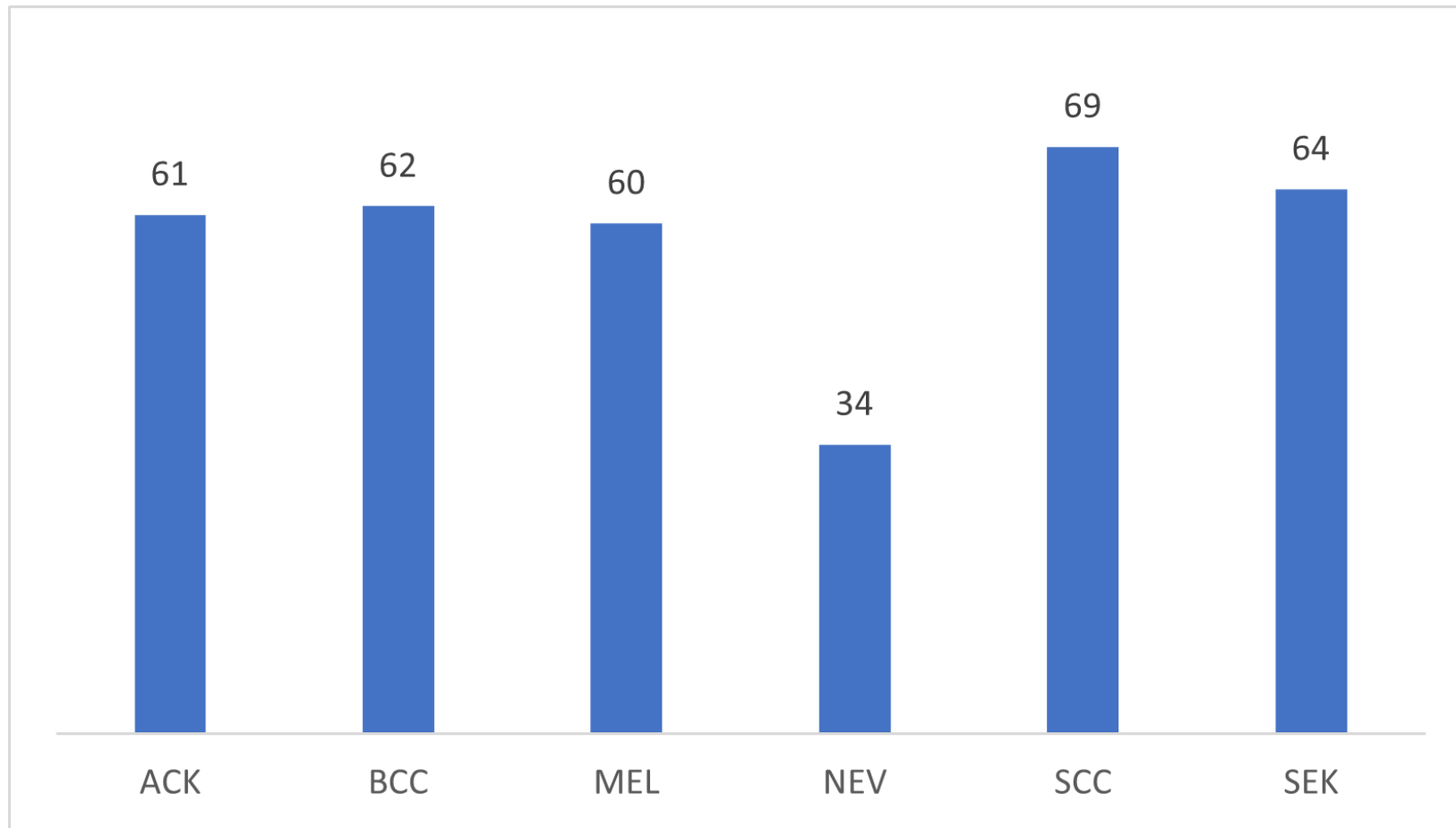
pgAdmin 4 (PostgreSQL)



METHODOLOGY

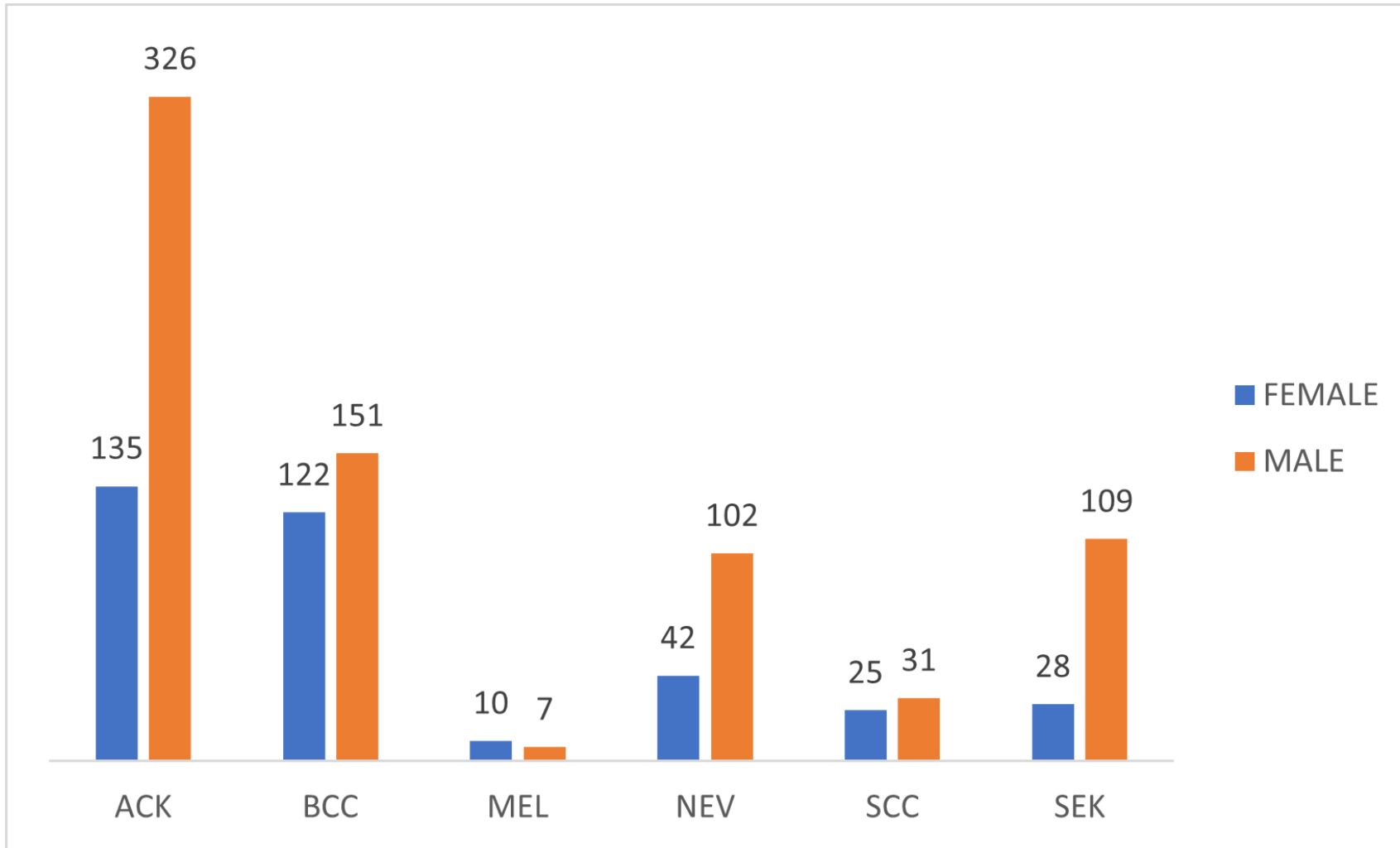


RESULTS



ACK: Actinic Keratosis
 BCC: Basal Cell Carcinoma
 MEL: Melanoma
 NEV: Nevus
 SCC: Squamous Cell Carcinoma
 SEK: Seborrheic Keratosis

Figure 1: Age Distribution Across Skin Lesion types



ACK: Actinic Keratosis
 BCC: Basal Cell Carcinoma
 MEL: Melanoma
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 SCC: Squamous Cell Carcinoma
 SEK: Seborrheic Keratosis

Figure 2: Gender Distribution Across Skin Lesion Type

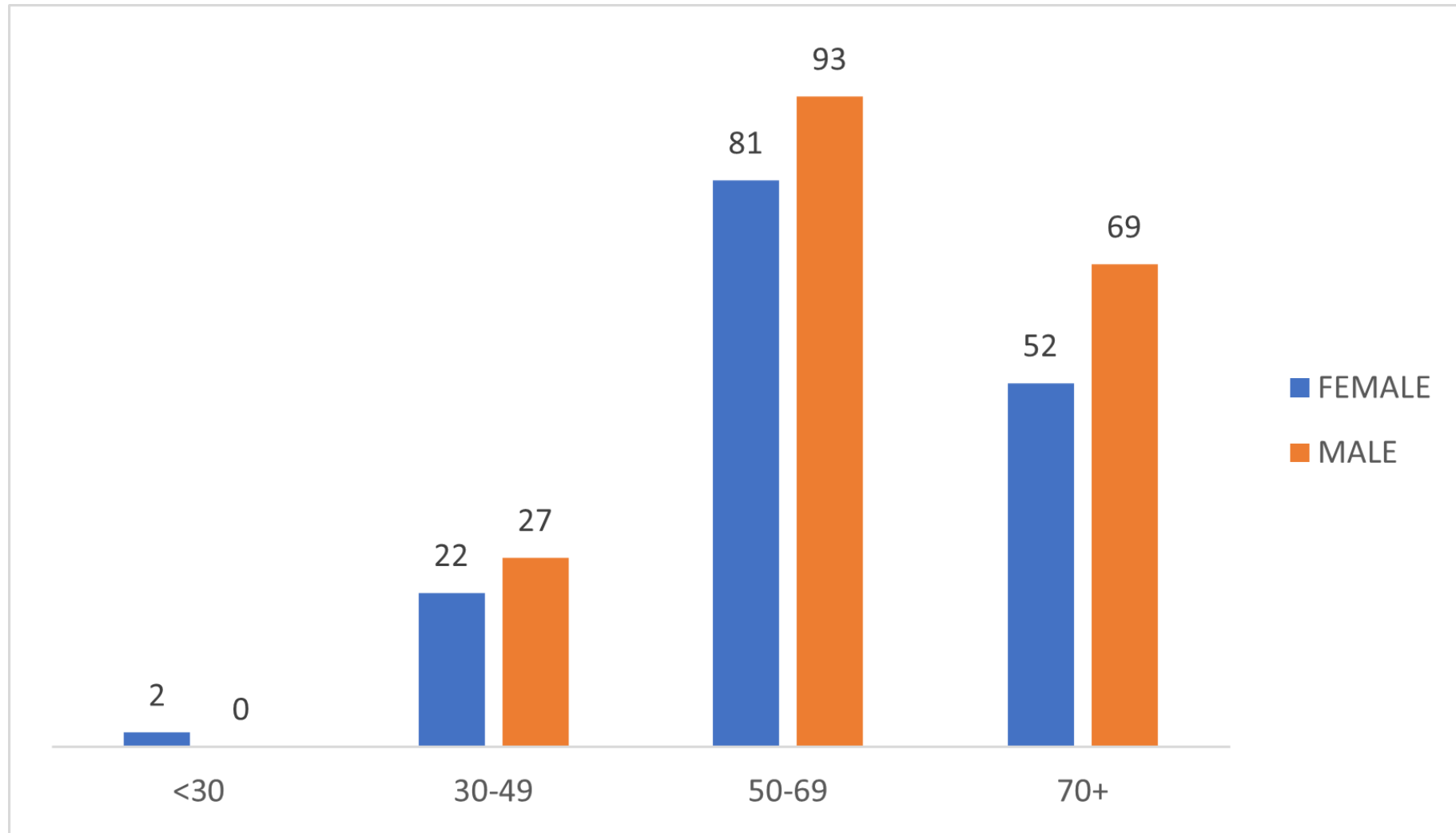


Figure 3: Age and Gender Distribution Of Malignant Lesions

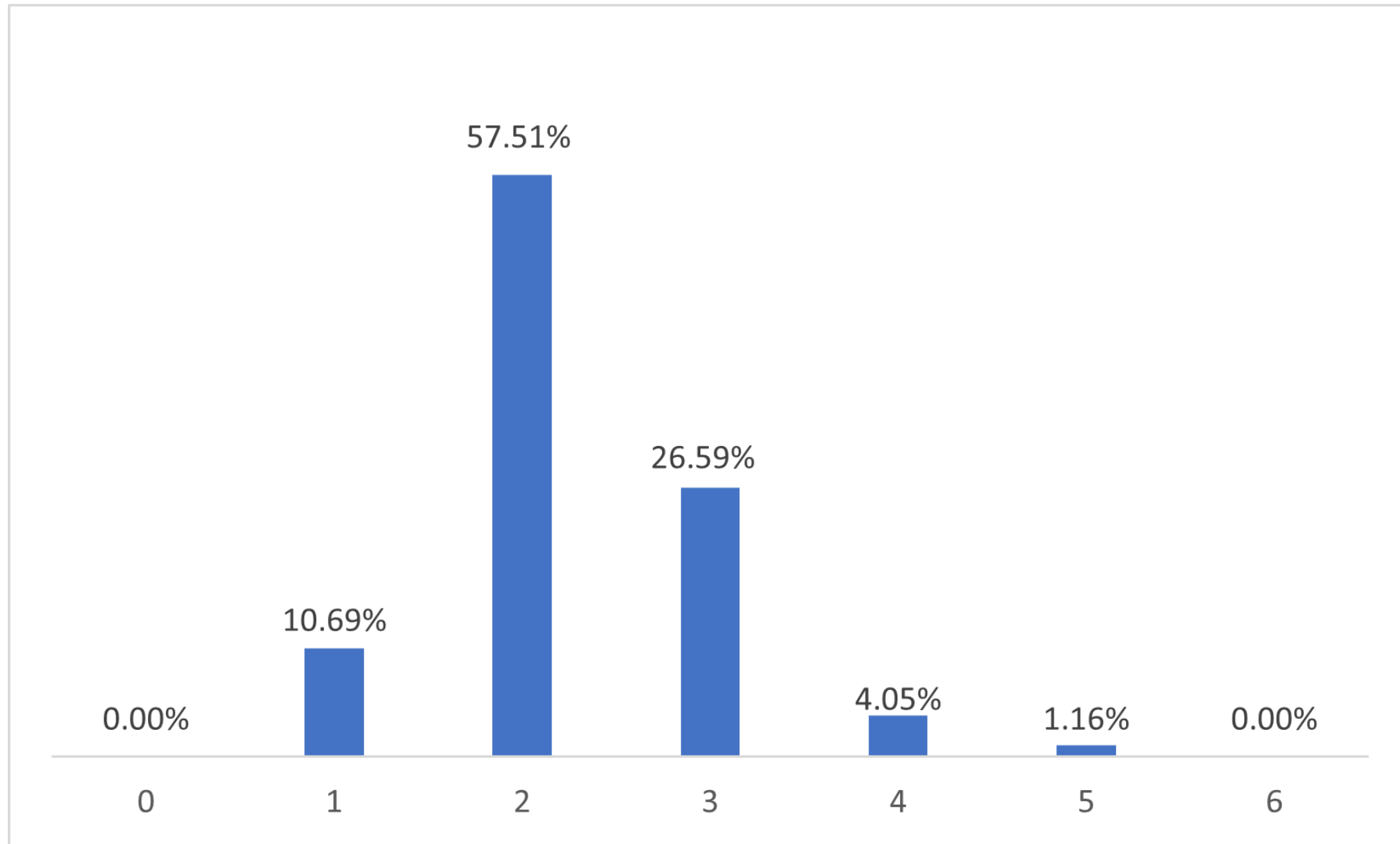


Figure 4: Percentage Distribution of Malignant Lesion by Fitzpatrick Skin Type

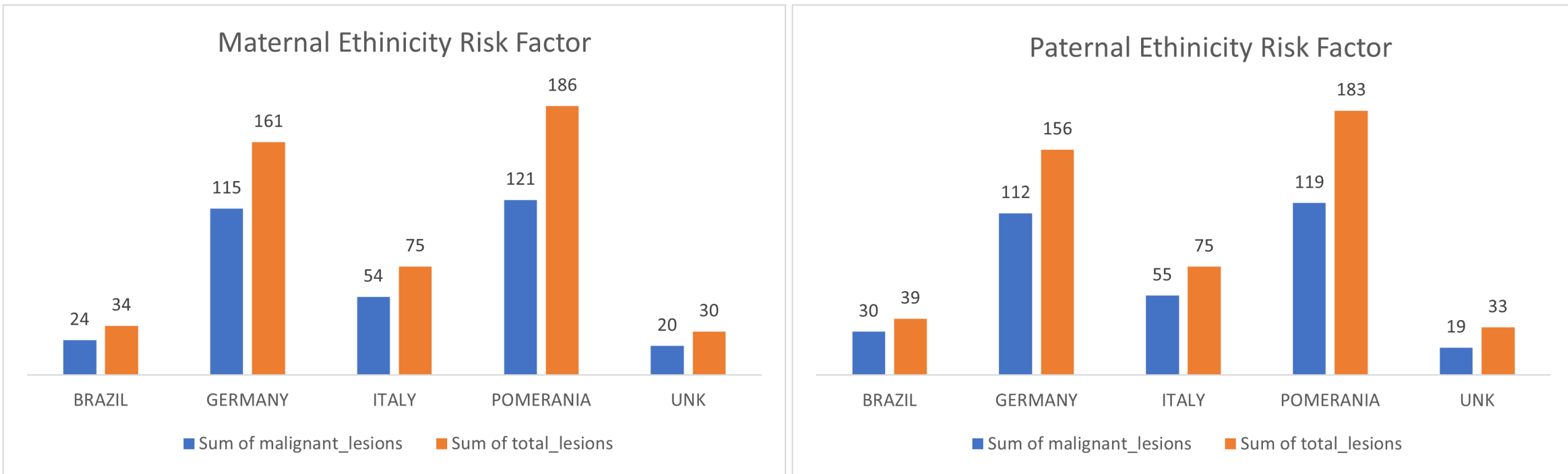
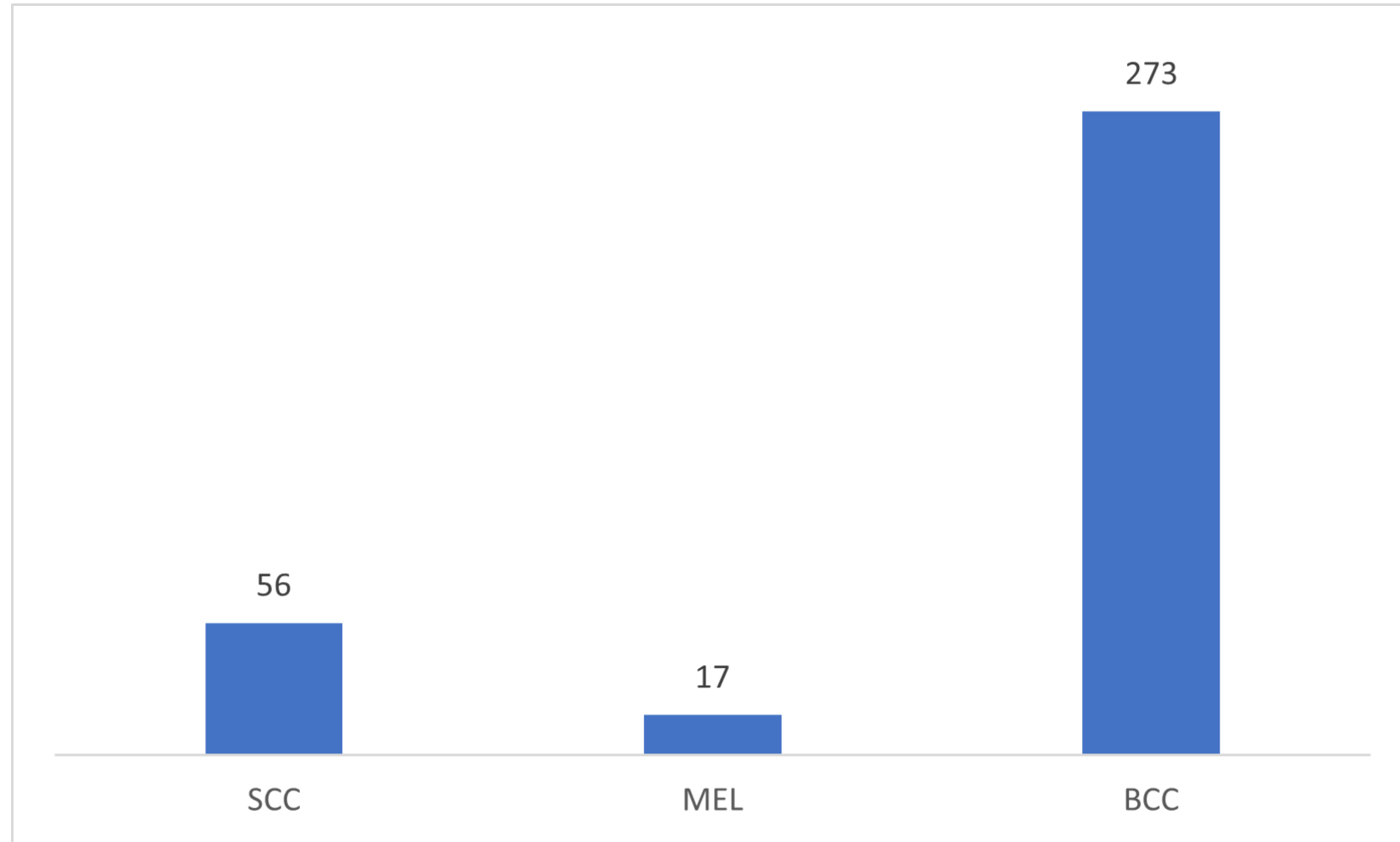


Figure 5: Background ethnicity distribution relationship with Skin Lesions



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Carcinoma
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Figure 6: Distriution of Malignant Lesions

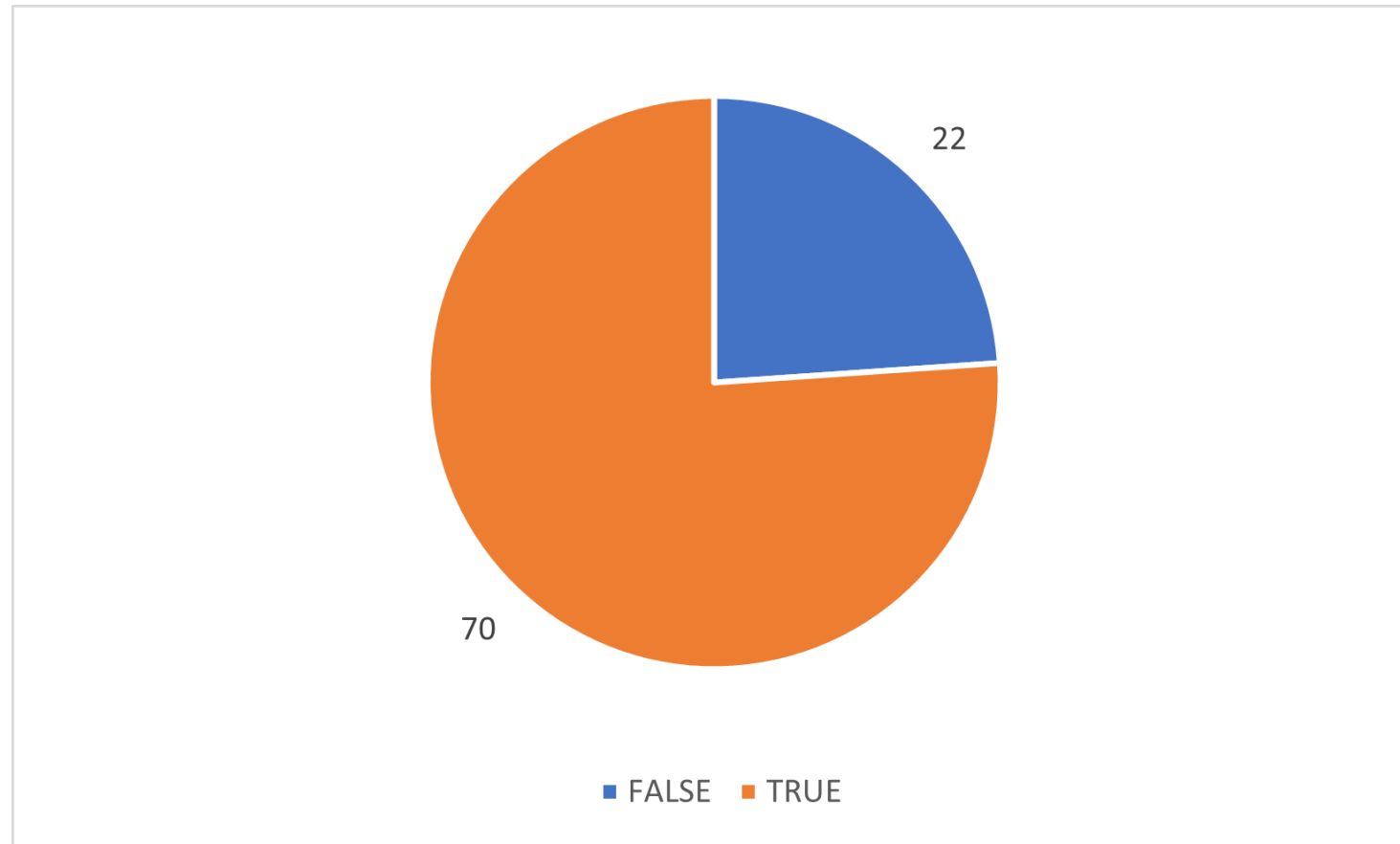


Figure 7:Relationship Between Pesticide Exposure and Skin Lesions

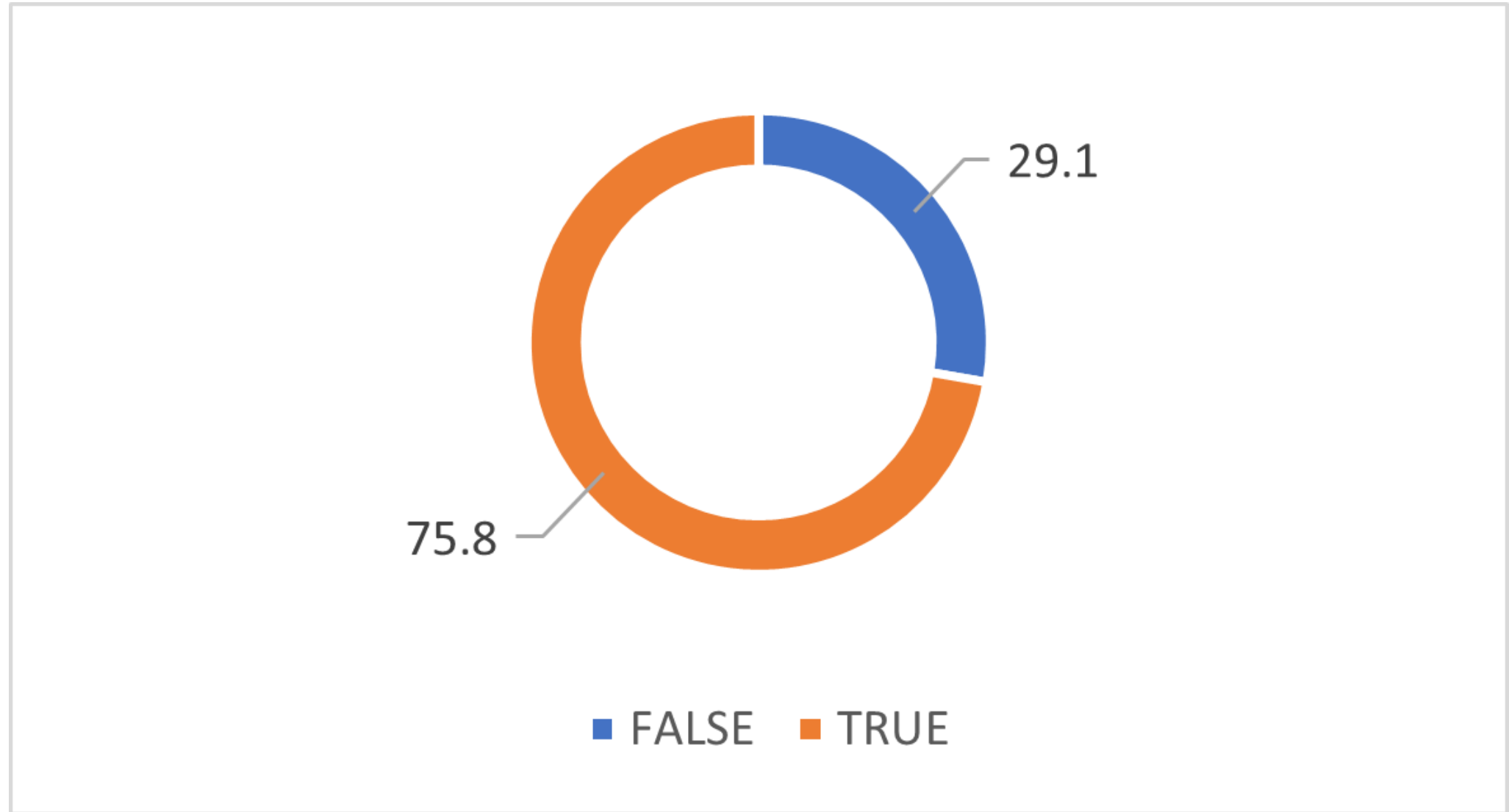


Figure 8: Relationship Between Smoking and Skin Lesions

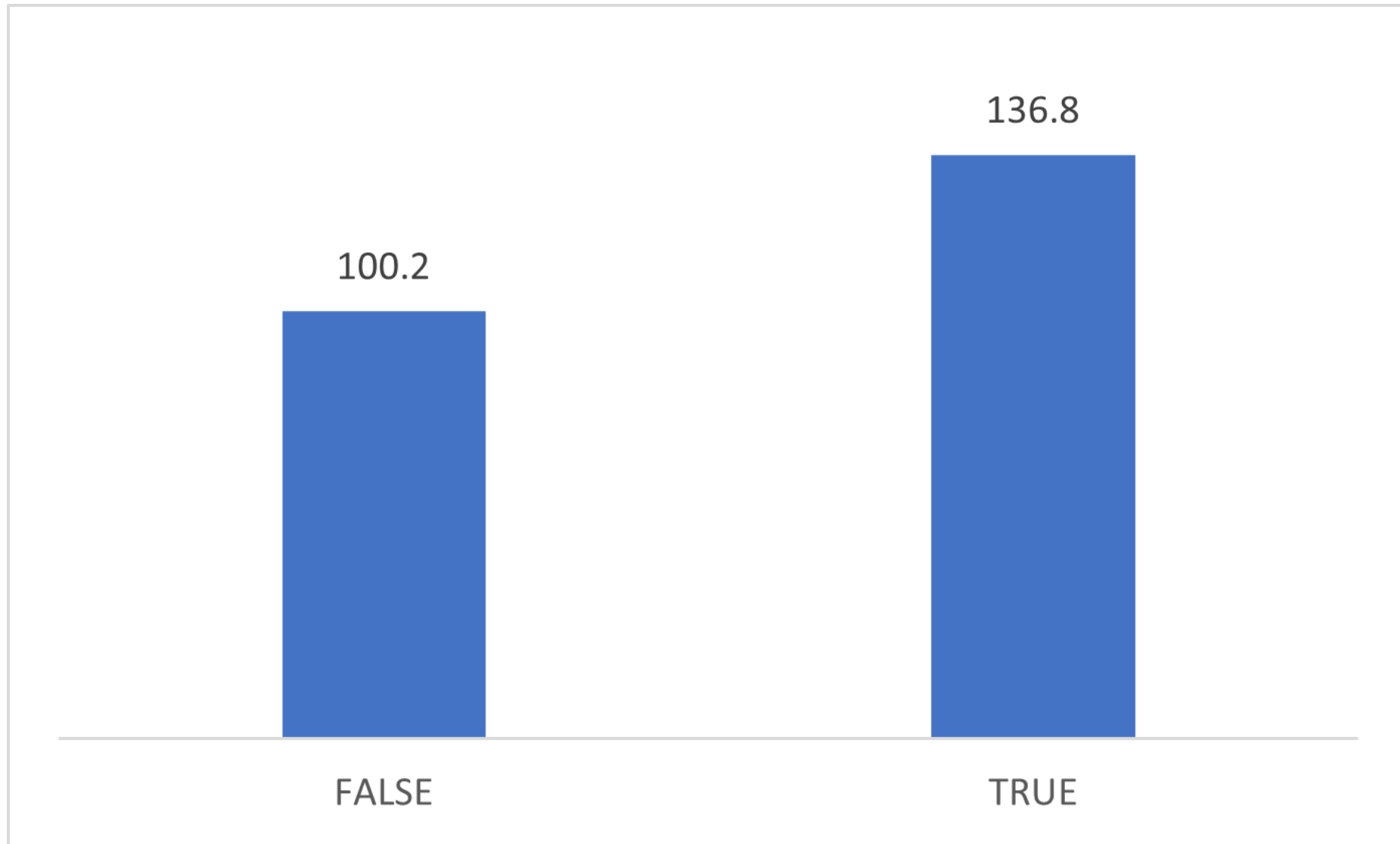


Figure 9: Pipe Water Predisposition To Malignancy

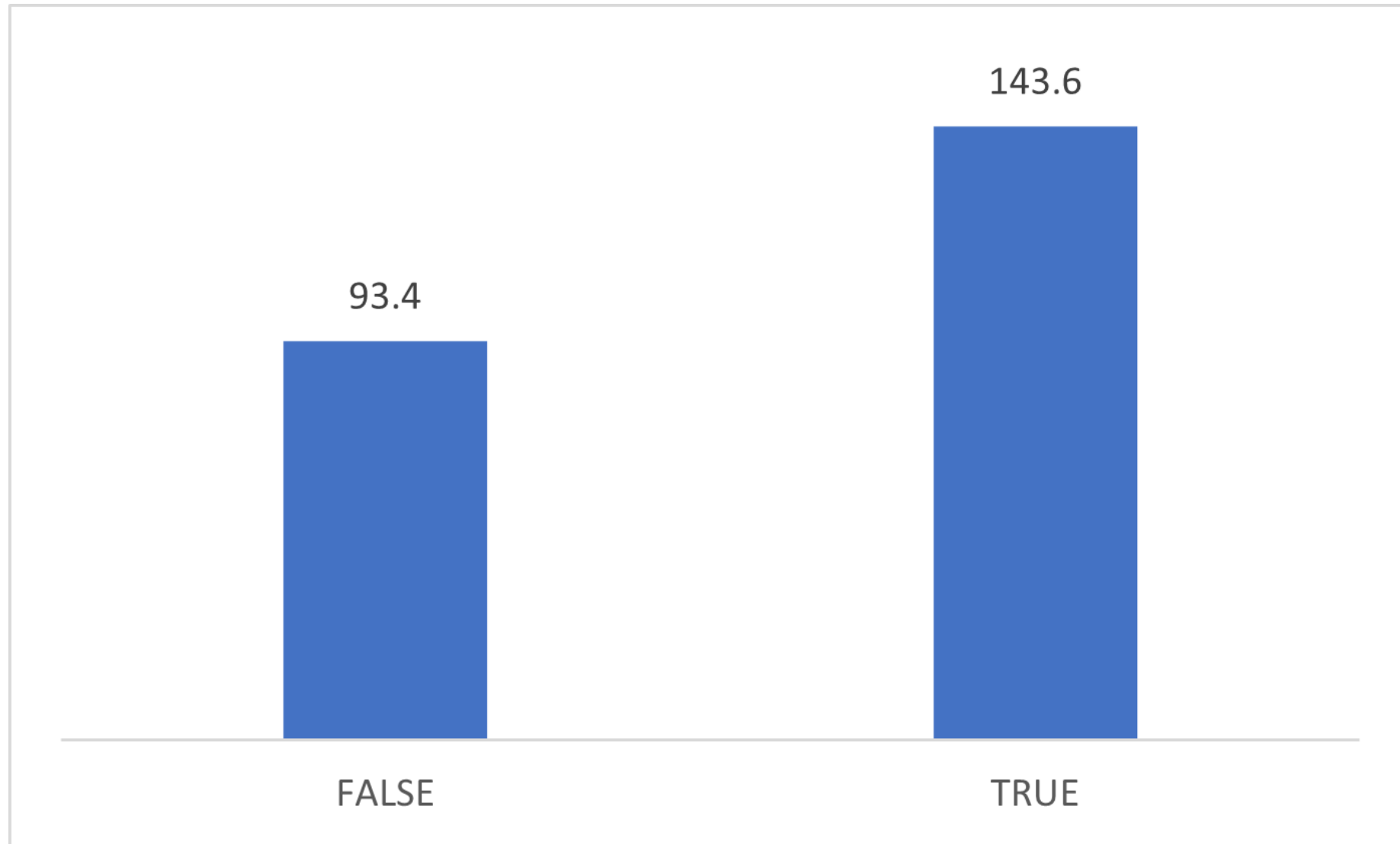


Figure 10: Risk of Malignancy in Relation to Sewage Water Exposure

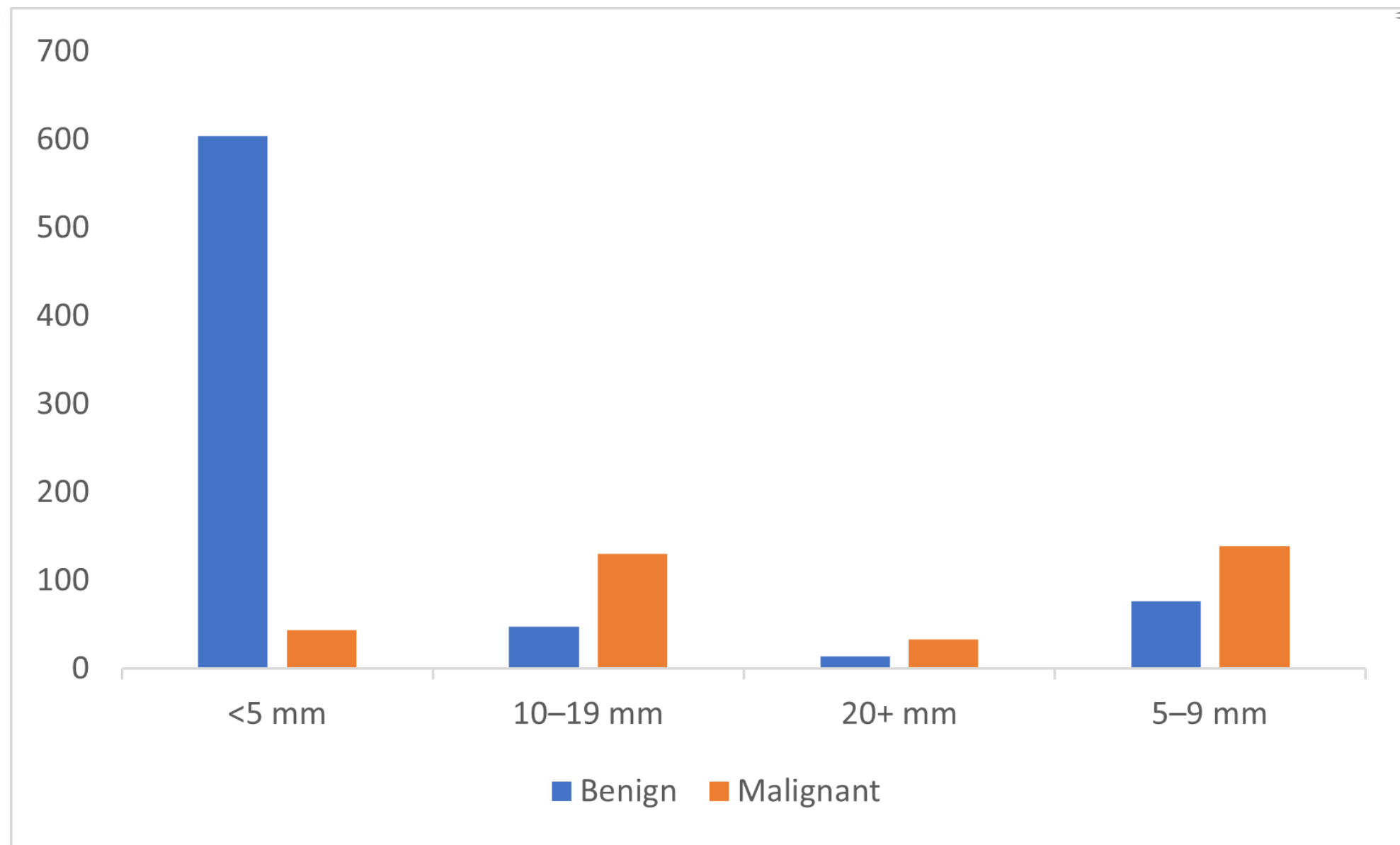


Figure 11: Size of Skin Lesion By Tumor Type

Fitzpatrick▼	Region▼	Mean Diameter▼	Itch▼	Grew▼	Hurt▼	Change▼	Bleed Elevation▼	Column1▼	Biopsied▼
0	FOREARM	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	FACE	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	ARM	0	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	ARM	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	CHEST	0	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	NOSE	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	FACE	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	FACE	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	FACE	0	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	NOSE	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	FOREARM	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	FACE	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	ARM	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	FACE	0	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	FOREARM	0	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
2	ARM	10.5	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE
0	HAND	0	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	FOREARM	0	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE
0	NOSE	0	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	FACE	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	CHEST	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	CHEST	0	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE
0	FOREARM	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	FOREARM	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
0	CHEST	0	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE

Figure 12: Machine Ready Metadata for Early Detection of Skin Cancer



DISCUSSION

- ❖ This queries support that malignant lesions are largely age-associated, while benign lesions are not
- ❖ Age is a strong demographic risk factor in malignant skin lesions
- ❖ Male gender is a significant demographic factor in lesion distribution
- ❖ Piped water and pesticide contains arsenic which is a naturally occurring toxin and human carcinogens, thus usage or exposure to such predisposes to development of malignant lesions
- ❖ Tumor size further serves as a strong clinical indicator of malignancy
- ❖ The insight shows malignant lesions are overwhelmingly concentrated in lighter Fitzpatrick types (especially Type II 57.5%)



RECOMMENDATION

- ❖ Preventive screening efforts should particularly target adults over 55–60 years.
- ❖ UV protection awareness should focus on high-risk skin types
- ❖ Public health interventions should particularly target older men
- ❖ Chronic environmental exposure (arsenic pesticide + possible water contamination) may contribute to increased non-melanoma skin cancer risk.
- ❖ So recommend preventive measures such as:
 - Reducing exposure
 - Early detection
 - Infrastructure safety
 - Education

Conclusion

- ❖ Skin cancer is one of the most common and life-threatening diseases
- ❖ misdiagnosis, lack of access to dermatologists, and limited understanding of environmental risk factors delay diagnosis and early treatment
- ❖ By analysing patient's demographics, risk factors and diseases characteristics, patterns are uncovered and structured into machine learning-ready dataset that supports early detection of skin cancer



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THANK YOU